

Trial Higher School Certificate 2011
Extension 2

Time: 3 hours**Total 120 marks****Question 1 (15 marks) (begin on a new page)****Marks**

a) Find $\int \frac{x}{\sqrt{16 - x^2}} dx$. 2

b) By completing the square, find $\int \frac{8}{x^2 + 4x + 13} dx$. 2

c) i) Find real numbers a, b, c , such that
$$\frac{7x + 4}{(x^2 + 1)(x + 2)} = \frac{ax + b}{x^2 + 1} + \frac{c}{x + 2} .$$
 3

ii) Hence, find $\int \frac{7x + 4}{(x^2 + 1)(x + 2)} dx$. 2

d) Use integration by parts to find $\int x^2 \log_e x \, dx$. 3

e) Use the substitution $u = \cos x$, to find $\int \cos^2 x \sin^7 x \, dx$. 3

Question 2 (15 marks) (begin on a new page)**Marks**a) Let $z = 2 + i$, $w = 1 - i$. Find, in the form $x + iy$,i) $3z + iw$, **1**ii) $\overline{z w}$, **1**iii) $\frac{5}{z}$. **1**b) Let $\alpha = -\sqrt{3} + i$.i) Express α in modulus - argument form. **2**ii) Express α^4 in modulus - argument form. **1**iii) Hence express α^4 in form $x + iy$. **1**

c) Sketch the region in the complex plane where the inequalities

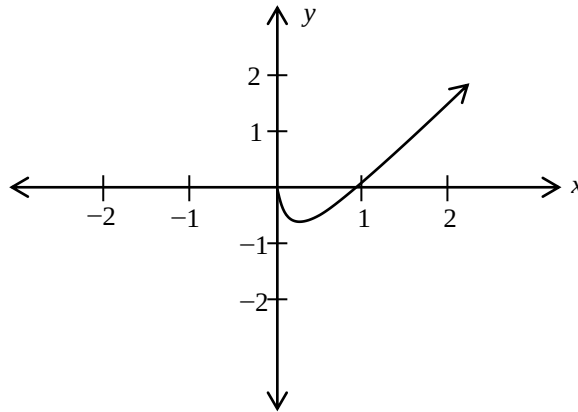
$$|z - i| \leq 2 \text{ and } 0 \leq \arg(z - 1) \leq \frac{3\pi}{4} \text{ hold simultaneously.}$$
3

d) Let $z_1 = 4 + i$ and $z_2 = 1 + 2i$. Let points A, B, C represent the complex numbers $z_1, z_2, z_1 - z_2$, respectively, on the complex plane.i) On a diagram, show the points A, B, C . Indicate any geometrical relationships on your diagram. **1**ii) The point A is rotated through 90° in the anticlockwise direction about B to the point D . Write down the complex number represented by D . **1**e) On an Argand diagram, sketch and describe geometrically the locus of z such that

$$|z| = |z - 4|.$$
3

Question 3 (15 marks) (begin on a new page)**Marks**

- a) The diagram below shows the graph of the function $y = f(x)$.



Draw separate, one-third page sketches of the graphs of the following:

- | | | |
|------|-----------------------|----------|
| i) | $y = -f(x),$ | 1 |
| ii) | $y = f(x) ,$ | 1 |
| iii) | $y = \frac{1}{f(x)},$ | 2 |
| iv) | $y = [f(x)]^2.$ | 2 |
- b) i) On the same set of axes, sketch the graphs of $y = \log_e x$ and $y = \frac{2}{x}.$ **1**
- ii) Hence, on a separate diagram, sketch the graph of $y = \frac{2\log_e x}{x}.$ **4**

Indicate on your graph any asymptotes and the co-ordinates of any stationary points.

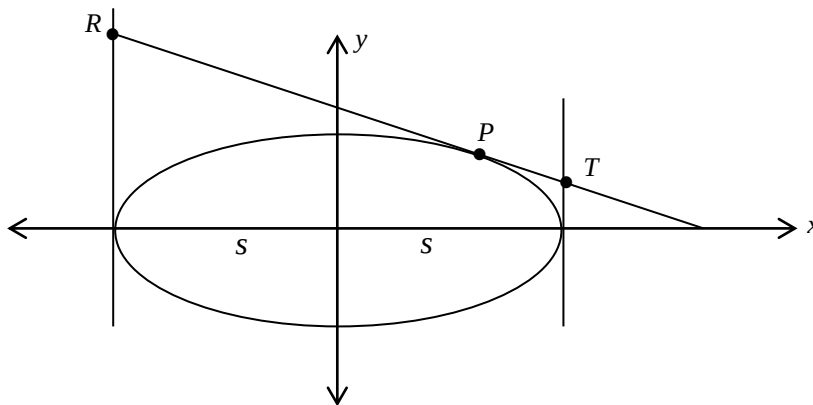
- c) Sketch the graph of $y^2 = (x - 2)(x - 3),$ including any asymptotes. **4**

Question 4 (15 marks) (begin on a new page)**Marks**

a) Consider the hyperbola with the equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

- i) What is the eccentricity of the hyperbola? 1
- ii) Find the co-ordinates of the foci and x intercepts of the hyperbola. 2
- iii) Find the equations of the directrices and the asymptotes for the hyperbola. 2

b)



The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b > 0$, as shown in the

diagram above. The points S and S' are the foci. The tangent at P meets the tangents at the ends of the major axis at R and T .

- i) Show that the equation of the tangent at P is given by

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1. \quad 2$$

- ii) Show that RT subtends a right angle at S . 3

- iii) Show that the points R, T, S, S' are concyclic. 1

c) Let $Q(x_0, y_0)$ be an external point of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (That is $\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} > 1$).

Show that the equation of the chord of contact of the tangents from the point

Q to the ellipse, is given by the equation $\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$. 4

Question 5 (15 marks) (begin on a new page)**Marks**

- a) When $P(x) = x^4 + ax^2 + bx$ is divided by $x^2 + 1$, the remainder is $x + 2$.
Find the values of a and b , given that these values are real. **3**
- b) The graph of $y = 2x^3 - 3x^2 - 12x + k$ has turning points at $x = 2$ and $x = -1$.
Find the values of k such that the equation $y = 2x^3 - 3x^2 - 12x + k = 0$ has three real and distinct roots. **2**
- c) A nine-member fund raising committee consists of four students, three teachers and two parents. The committee meets around a circular table, such that all the students sit together as a group, all the teachers sit together as another group, but no teacher sits next to a student.
- i) How many different arrangements are possible for the members of the committee to sit around the table. **2**
- ii) One student and one parent are related. Given that all arrangements in b i) are equally likely, what is the probability that these two members sit next to each other? **2**
- d) i) Write down the three relations which hold between roots α, β, γ of the equation $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$ and the coefficients a, b, c, d . **1**
- ii) Consider the equation $36x^3 - 12x^2 - 11x + 2 = 0$.
You are given that the roots α, β, γ of this equation satisfy $\alpha = \beta + \gamma$.
Use part i) to find α . **2**
- iii) Suppose the equation $x^3 + px^2 + qx + r = 0$ has roots α, β, γ , which satisfy $\alpha = \beta + \gamma$.
Show that $p^3 - 4pq + 8r = 0$. **3**

Question 6 (15 marks) (begin on a new page)**Marks**

a)

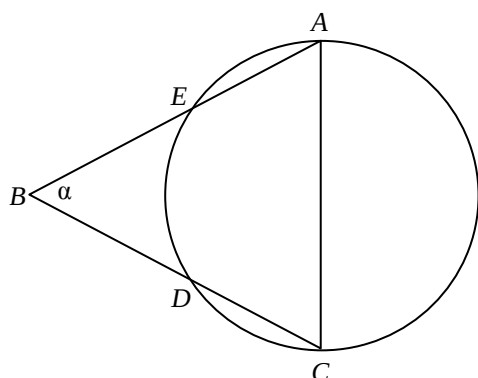


Diagram not to scale

In the diagram above, AC is a diameter of a circle with the point B outside the circle. The intervals BC and BA meet the circle in the points D and E respectively.

Also, $AC = BC$. Let $BA = x$, $BC = y$ and $\angle ABC = \alpha$.

i) Show that $\cos \alpha = \frac{x}{2y}$. 2

ii) Find the length DC in terms of x and y . 4

b) Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n t \, dt$, where $n \geq 0$ is an integer.

i) Show that $I_n = \left(\frac{n-1}{n} \right) I_{n-2}$ for $n \geq 2$. 2

ii) Hence find the exact value of I_4 . 2

Question 6 continues on the next page.

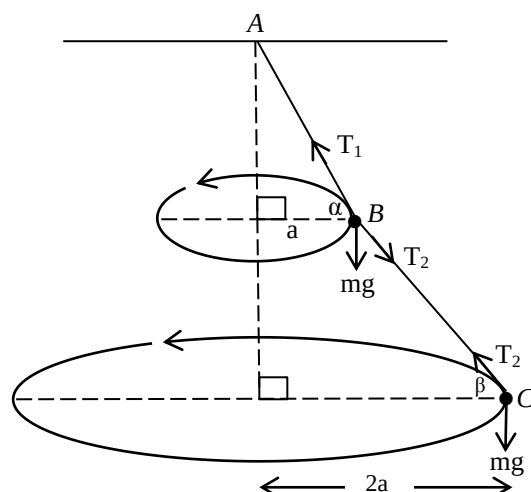
- | | | | |
|-----|-----|--|--------------|
| 6c) | i) | Let $k > 0$ be a real number. | Marks |
| | | <p>If $\frac{(k+1)^{k-1}}{k^k} < 1$, show that $(k+1)^{k+1} > \frac{(k+1)^{2k}}{k^k}$.</p> | 2 |
| | ii) | <p>Prove, by mathematical induction, for all integers $n \geq 2$ that</p> <p>$n^n > (n+1)^{n-1}$.</p> | 3 |

Question 7 (15 marks) (begin on a new page)

- a) Let $I_n = \int \frac{dx}{(x^2+1)^n}$ where $n \geq 1$, is an integer.
- i) Show that, for $n \geq 2$, $I_n = \frac{1}{2(n-1)} \left[\frac{x}{(x^2+1)^{n-1}} + (2n-3)I_{n-1} \right]$. **4**
- ii) Hence evaluate $\int_0^1 \frac{dx}{(x^2+1)^2}$. **2**

Question 7 continues on the next page.

7b)

**Diagram****not to****scale**

A light, inextensible string ABC where $AB = \frac{5a}{3}$ and is inclined at an angle of α to the horizontal, while $BC = \frac{5a}{4}$ and is inclined at an angle β to the horizontal.

At B is attached a particle of mass $7m$ and at C is attached a particle of mass m .

The end A is attached to a fixed point and the whole system rotates steadily with uniform angular velocity about the vertical through A in such a way that B and C describe horizontal circles of radii a and $2a$ respectively.

The tensions in the strings AB and BC are T_1 and T_2 respectively. The strings remain taut.

The acceleration due to gravity is g .

- | | | |
|------|--|----------|
| i) | Show that $T_2 = \frac{5}{3}mg$. | 3 |
| ii) | Find T_1 . | 2 |
| iii) | If v_1 is the speed of B and v_2 is the speed of C , find the value of $\frac{v_1}{v_2}$. | 4 |

Question 8 (15 marks) (begin on a new page)

Marks

- a) The region between the curve $y = \sin x$ and the line $y = 1$, from $x = 0$ to $x = \frac{\pi}{2}$, is rotated around the line $y = 1$.

Using a slicing technique, find the exact volume formed.

4

- b) i) Differentiate with respect to x , the function, $h(x)$ given by

$$h(x) = \frac{\log_e x}{x} \text{ for } x > 0.$$

1

- ii) Given that the only stationary point of $y = h(x)$ is a maximum turning point, deduce, without calculating any numerical values, that $e^\pi > \pi^e$.

[you may assume that $\pi > e$]

2

- c) The diagram shows a boat showroom built on level ground. The length of showroom is 100 m. At one end of the showroom, the shape is a square measuring 20 m by 20 m and at the other end an isosceles triangle of height 20 m and base 10 m. The trapezium $ABCD$ is a cross section of the showroom taken parallel to the ends.

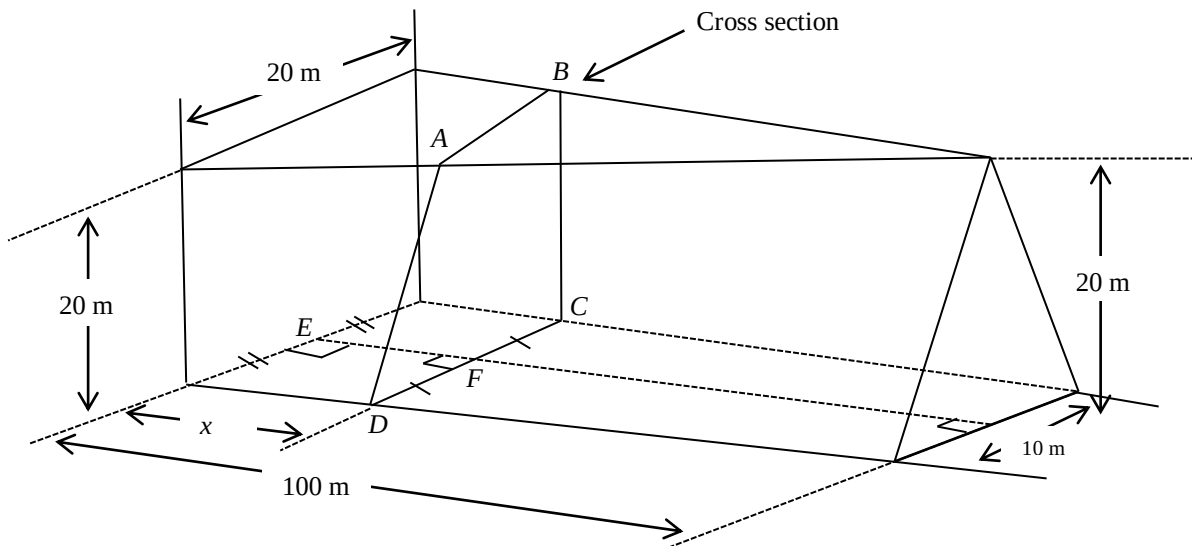


Diagram not to scale.

- i) If EF is x metres in length, show that the length of DC is $\left(20 - \frac{x}{10}\right)$ metres.

2

- ii) By considering trapezoidal slices parallel to the ends of the showroom, find the volume enclosed by the showroom.

6

END OF EXAMINATION

Question 1

(a)(i) Let $u = 16 - x^2$ then $\frac{du}{dx} = -2x$

$$x dx = -\frac{1}{2} du$$

$$\begin{aligned} \int \frac{x}{\sqrt{16-x^2}} dx &= -\frac{1}{2} \int u^{-\frac{1}{2}} du \\ &= -\frac{1}{2} \times 2u^{\frac{1}{2}} \\ &= -\sqrt{16-x^2} \end{aligned}$$

(b)
$$\int \frac{8}{x^2 + 4x + 13} dx = 8 \int \frac{dx}{(x+2)^2 + 3^2}$$

$$= \frac{8}{3} \tan^{-1} \frac{(x+2)}{3} + c$$

(c)(i)
$$\frac{7x+4}{(x^2+1)(x+2)} = \frac{ax+b}{x^2+1} + \frac{c}{x+2}$$

$$7x+4 = (ax+b)(x+2) + c(x^2+1)$$

Let $x = -2$ and $x = 0$

$$-10 = 5c$$

$$4 = b(0+2) - 2(0^2+1)$$

$$c = -2$$

$$b = 3$$

Equating the coefficients of x^2 , $0 = a - 2$

$$a = 2$$

$\therefore a = 2, b = 3$ and $c = -2$

(c)(ii)
$$\begin{aligned} \int \frac{7x+4}{(x^2+1)(x+2)} dx &= \int \left(\frac{2x+3}{(x^2+1)} - \frac{2}{(x+2)} \right) dx \\ &= \int \left(\frac{2x}{(x^2+1)} + \frac{3}{(x^2+1)} - \frac{2}{(x+2)} \right) dx \\ &= \ln(x^2+1) + 3 \tan^{-1} x - 2 \ln |x+2| + c \end{aligned}$$

Q1

$$\begin{aligned}
 1) \quad \int x^2 \log_e x \, dx &= \log_e x \times \frac{x^3}{3} - \int \frac{x^3}{3} \frac{1}{x} dx \\
 &= \frac{x^3 \log_e x}{3} - \frac{1}{3} \int x^2 dx \\
 &= \frac{x^3 \log_e x}{3} - \frac{x^3}{9} + c
 \end{aligned}$$

e) Let $u = \cos x$ then $-du = \sin x dx$

Now $\sin^6 x = (\sin^2 x)^3$

$$\begin{aligned}
 &= (1 - \cos^2 x)^3 \\
 &= 1 - 3\cos^2 x + 3\cos^4 x - \cos^6 x
 \end{aligned}$$

$$\begin{aligned}
 \int \cos^2 x \sin^7 x \, dx &= \int \cos^2 x \sin^6 x \sin x \, dx \\
 &= \int u^2 (1 - 3u^2 + 3u^4 - u^6) (-du) \\
 &= \int -u^2 + 3u^4 - 3u^6 + u^8 \, du \\
 &= -\frac{u^3}{3} + \frac{3u^5}{5} - \frac{3u^7}{7} + \frac{u^9}{9} + c \\
 &= -\frac{\cos^3 x}{3} + \frac{3\cos^5 x}{5} - \frac{3\cos^7 x}{7} + \frac{\cos^9 x}{9} + c
 \end{aligned}$$

Question 2

(a)(i) $3z + iw = 3(2+i) + i(1-i)$

$$\begin{aligned}
 &= 6 + 3i + i - i^2 \\
 &= 7 + 4i
 \end{aligned}$$

(a)(ii) $z\bar{w} = (2+i)(1+i)$

$$\begin{aligned}
 &= 2 + 2i + i + i^2 \\
 &= 1 + 3i
 \end{aligned}$$

(a)(iii) $\frac{5}{z} = \frac{5}{2+i} \times \frac{2-i}{2-i}$

$$\begin{aligned}
 &= \frac{5(2-i)}{4+1} \\
 &= 2-i
 \end{aligned}$$

Q.2

(b)(i)

Angle α is in the 2nd quadrant with $\arg \alpha = \frac{5\pi}{6}$ and

$$(|\alpha|)^2 = (\sqrt{3})^2 + 1^2 \quad \tan \alpha = -\frac{1}{\sqrt{3}}$$

$$= 3 + 1$$

$$|\alpha| = 2$$

Modulus-argument form $2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$

(b)(ii)

De Moivre's theorem

$$\alpha^4 = (2 \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})^4$$

$$= 2^4 (\cos 4 \times \frac{5\pi}{6} + i \sin 4 \times \frac{5\pi}{6})$$

$$= 16 (\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3})$$

$$= 16 (\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3})$$

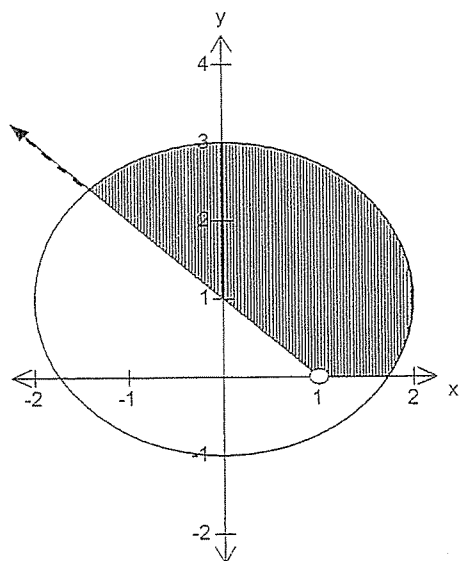
(b)(iii)

$$\alpha^4 = 16 (\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3})$$

$$= 16 (-\frac{1}{2} + i \times -\frac{\sqrt{3}}{2})$$

$$= -8 - 8\sqrt{3}i$$

(c)

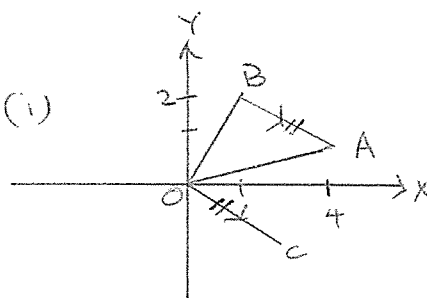


$|z - i| \leq 2$ represents a region with a centre is $(0, 1)$ and radius is less than or equal to 2.

$0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$ represents a region between angle 0

and $\frac{3\pi}{4}$ whose vertex is $(1, 0)$, not including the vertex

2(d)(i)



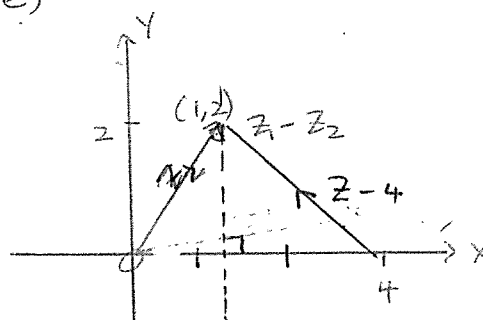
(ii) D rep

$$z_2 + i(z_1 - z_2)$$

$$= 1 + 2i + i(3 - i)$$

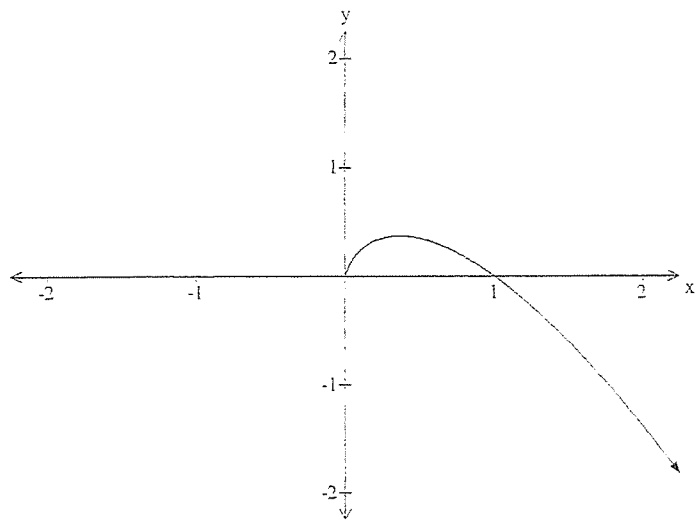
$$= 2 + 5i$$

7e)

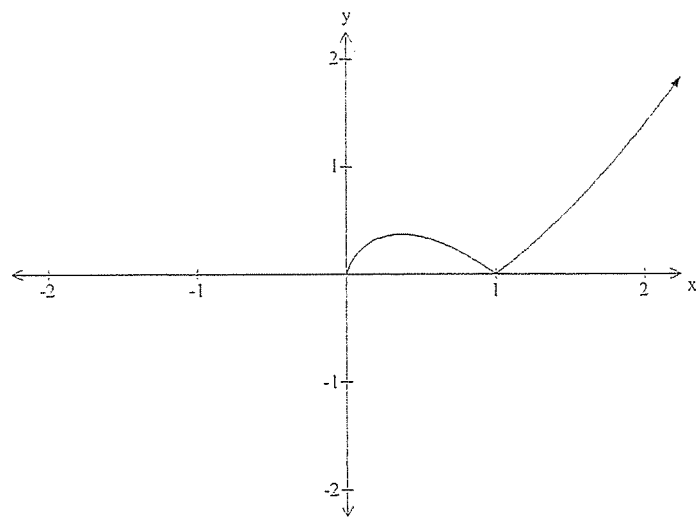


Question 3

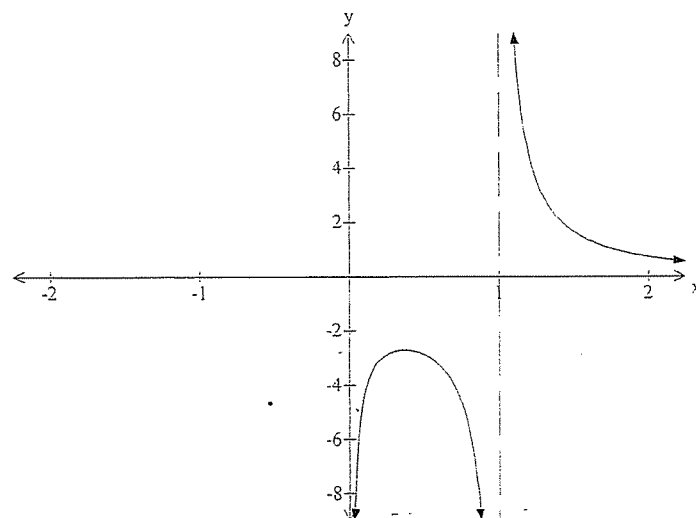
a)(i)



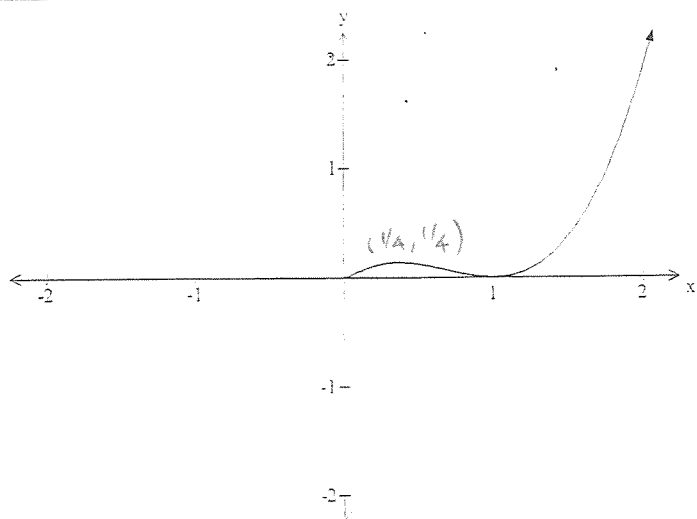
a)(ii)



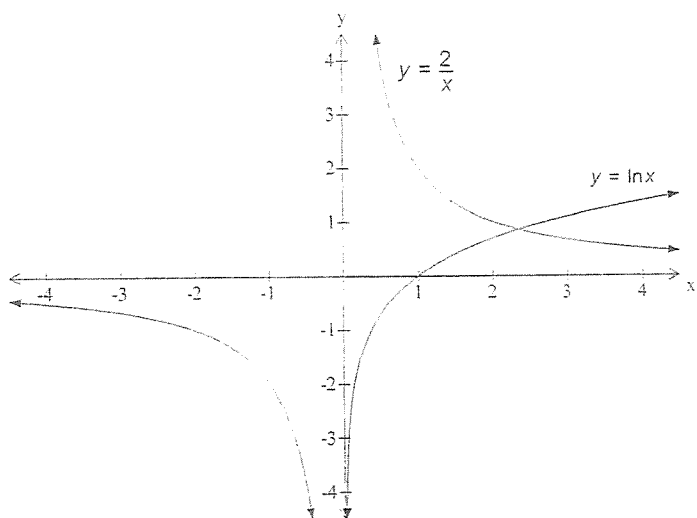
a)(iii)



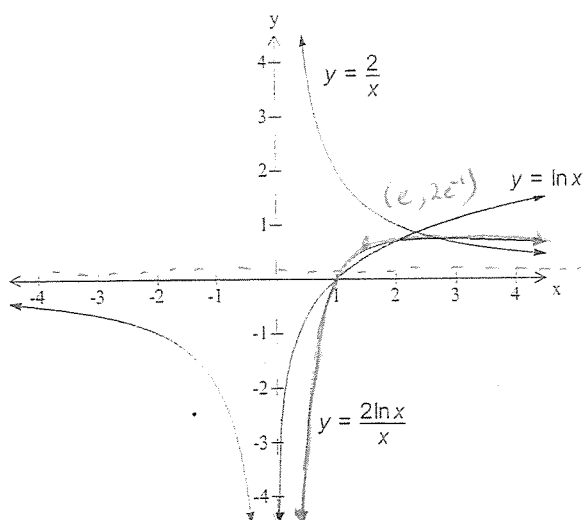
(a)(iv)



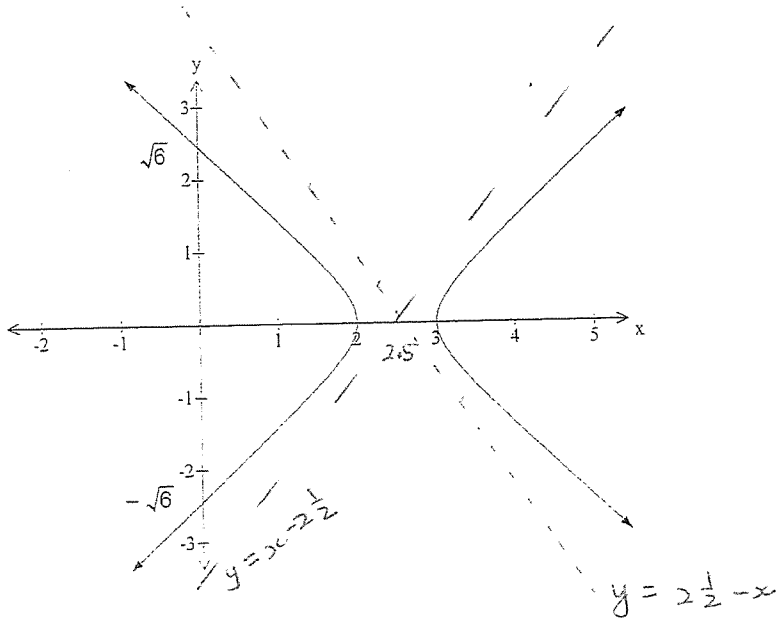
(b)(i)



(b)(ii)



Q3
c)



Question 4

(a)(i)

Hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ has $a^2 = 16$ and $b^2 = 9$.
 $a = 4$ $b = 3$

Eccentricity $b^2 = a^2(e^2 - 1)$
 $9 = 16(e^2 - 1)$

$$(e^2 - 1) = \frac{9}{16}$$

$$e^2 = \frac{25}{16}$$

$$e = \frac{5}{4}$$

(a)(ii)

Foci of a hyperbola are $(\pm ae, 0)$.

Foci are $(\pm 5, 0)$

X-intercepts of a hyperbola are ± 4 .

The points are $(\pm 4, 0)$

(a)(iii)

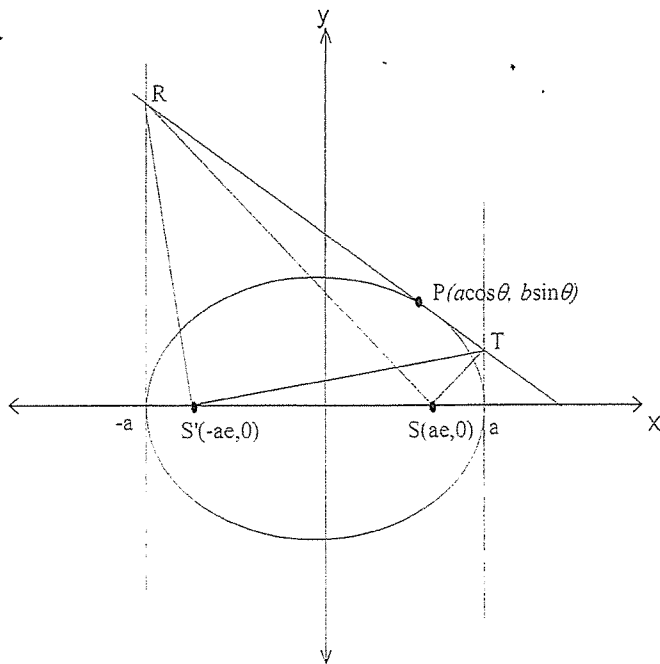
Directrices of a hyperbola are $x = \pm \frac{a}{e}$.

Directrices are $x = \pm \frac{16}{5}$

Asymptotes of a hyperbola are $y = \pm \frac{b}{a}x$.

Asymptotes are $y = \pm \frac{3}{4}x$

b) Q 4



b)(i) To find the equation of tangent through P

$$x = a \cos \theta$$

$$y = b \sin \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta$$

$$\frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= b \cos \theta \times \frac{1}{-a \sin \theta}$$

$$= \frac{-b \cos \theta}{a \sin \theta}$$

Equation of the tangent

$$y - y_1 = m(x - x_1)$$

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab(\sin^2 \theta + \cos^2 \theta)$$

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

(b)(ii) Q 4
At T $x = a$ then $\frac{a}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$

$$\frac{y}{b} \sin \theta = 1 - \cos \theta$$

$$y = \frac{b(1 - \cos \theta)}{\sin \theta}$$

At R $x = -a$ then similarly $y = \frac{b(1 + \cos \theta)}{\sin \theta}$

Gradients of lines at the focus $S(ae, 0)$

Gradient RS $\times$ Gradient TS

$$= \frac{\frac{b(1 + \cos \theta)}{\sin \theta} - 0}{-a - ae} \times \frac{\frac{b(1 - \cos \theta)}{\sin \theta} - 0}{a - ae}$$

$$= \frac{b(1 + \cos \theta)}{-a(1 + e) \sin \theta} \times \frac{b(1 - \cos \theta)}{a(1 - e) \sin \theta}$$

$$= \frac{b^2(1 - \cos^2 \theta)}{-a^2(1 - e^2) \sin^2 \theta}$$

$$= -1$$

$$\therefore \angle RST = 90^\circ$$

(iii) If the diagram is reflected in the y -axis, the similar result follows for S' .

$\therefore R, T, S, S'$ are concyclic (opp $\angle$ s suppl).

OR

$$M_{TS} \times M_{RS} = -\frac{b^2}{a^2} \left(\frac{1 - \cos^2 \theta}{(1 - e^2) \sin^2 \theta} \right)$$

change $S \rightarrow S'$ means change a^2 to $(-a)^2 = a^2$

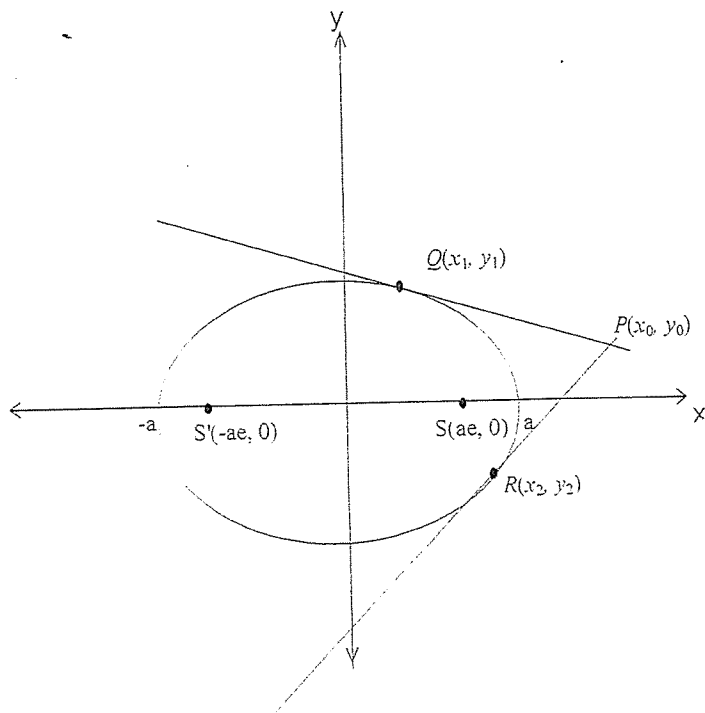
$$\text{Hence } M_{TS} \times M_{RS} = M_{TS'} \times M_{RS'} = -1$$

$\therefore$ As $\angle RS'T = \angle RST$, then

$RTSS'$ are concyclic as by

result ("angles subtended on same side of RT are equal")

4c)



To find the gradient of the tangent

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-2x}{a^2} \times \frac{b^2}{2y}$$

$$= \frac{-xb^2}{ya^2}$$

Equation of the tangent at $Q(x_1, y_1)$

$$y - y_1 = m(x - x_1)$$

$$y - y_1 = \frac{-x_1 b^2}{y_1 a^2} (x - x_1)$$

$$yy_1 a^2 - y_1^2 a^2 = -x_1 x b^2 + x_1^2 b^2$$

$$x_1 x b^2 + yy_1 a^2 = y_1^2 a^2 + x_1^2 b^2$$

$$\frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = \frac{y_1^2}{b^2} + \frac{x_1^2}{a^2}$$

$$\frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = 1$$

$P(x_0, y_0)$ is on the tangent at Q.

$$\therefore \frac{x_1 x_0}{a^2} + \frac{y_1 y_0}{b^2} = 1 \quad \text{Eqn 1}$$

Similarly the equation of the tangent at $R(x_2, y_2)$ is

$$\frac{x_2 x}{a^2} + \frac{y_2 y}{b^2} = 1$$

$P(x_0, y_0)$ is on the tangent at R

$$\therefore \frac{x_2 x_0}{a^2} + \frac{y_2 y_0}{b^2} = 1 \quad \text{Eqn 2}$$

Hence the points P & Q both satisfy

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$$

$\therefore$ The equation of the chord of

$$\text{contact } PQ \text{ is } \frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$$

Question 5

a) $P(x) = x^4 + ax^2 + bx$

$$x^2 + 1 = (x+i)(x-i)$$

$$P(i) = i^4 - a + bi$$

but $P(i) = i + 2$ From $P(x) = (x^2 + 1)q(x) + x + 2$

$$\therefore 1 - a = 2 \quad bi = i$$

$$a = -1, \quad b = 1$$

b) $y = 2x^3 - 3x^2 - 12x + k$

need $y(2) \times y(-1) < 0$

$$(2 \times 8 - 3 \times 4 - 24 + k)(-2 - 3 + 12 + k) < 0$$

$$(k - 20)(k + 7) < 0$$

$$-7 < k < 20$$

ci) 4! students and 3! teachers, $\therefore P_1 P_2 = 2 \times 3! \times 4! = 288$

cii) $S_1 P_1$ represents student, parent. S', P are remaining set of students and other parents, T is teachers.

$\therefore 3! =$ students and $3! =$ teachers.

$$2 \times 3! \times 3! = 72$$

$$\therefore \text{Prob} = \frac{72}{288} = \frac{1}{4}$$

$$5di) \alpha + \beta + \gamma = \frac{-b}{a} \quad \alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} \quad \alpha\beta\gamma = \frac{-d}{a}$$

$$dii) \alpha + \beta + \gamma = 2\alpha = \frac{12}{36} = \frac{1}{3}, \quad \alpha = \frac{1}{6}$$

$$diii) \alpha + \beta + \gamma = 2(\beta + \gamma) = -p$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \beta\gamma + (\beta + \gamma)\alpha$$

$$= \beta\gamma + (\beta + \gamma)^2$$

$$= (\beta^2 + 3\beta\gamma + \gamma^2)$$

$$= q$$

$$\alpha\beta\gamma = \beta\gamma + (\beta + \gamma) = -r$$

$$\therefore p^3 - 4pq + 8r$$

$$= -8(\beta + \gamma)^3 + 8(\beta + \gamma) + (\beta^2 + 3\beta\gamma + \gamma^2) - 8(\beta + \gamma)\beta\gamma$$

$$= -8(\beta + \gamma) \left[(\beta + \gamma)^2 - (\beta^2 + 3\beta\gamma + \gamma^2) + \beta\gamma \right]$$

$$= 0$$

Question 6

Assume $AB = x$

ai) join C to E , $\angle AEC = 90^\circ$

CE is the axis of symmetry for $\triangle ABC$.

$$EB = \frac{x}{2}$$

$$\therefore \cos \alpha = \frac{EB}{BC} = \frac{\frac{x}{2}}{y} = \frac{x}{2y}.$$

ii) join A to D , $\angle ADC = 90^\circ$

$$\cos \alpha = \frac{BD}{AB}$$

$$\therefore BD = AB \cos \alpha$$

$$= x \cdot \frac{x}{2y} = \frac{x^2}{2y}$$

$$\begin{aligned} \therefore DC &= y - \frac{x^2}{2y} \\ &= \frac{2y^2 - x^2}{2y} \end{aligned}$$

$$bi) I_n = \int_0^{\frac{\pi}{2}} \cos^n t \, dt$$

$$= \left[\sin t \cdot \cos^{n-1} t \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^2 t \cdot \cos^{n-2} t \, dt$$

$$I_n = (n-1) \int_0^{\frac{\pi}{2}} (1 - \cos^2 t) \cdot \cos^{n-2} t \, dt$$

$$= (n-1) [I_{n-2} - I_n]$$

$$= (n-1) I_{n-2} - n I_n + I_n$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{(n-1)}{n} I_{n-2}, \text{ for } n \geq 2$$

$$bii) I_4 = \frac{3}{4} I_2$$

$$= \frac{3 \cdot 1}{4 \cdot 2} I_0$$

$$\int_0^{\frac{\pi}{2}} dt = \left[t \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2}$$

$$I_4 = \frac{3}{8} + \frac{\pi}{2}$$

$$= \frac{3\pi}{16}$$

$$6ci) \quad 0 < \frac{(k+1)^{k-1}}{k^k} < 1$$

$$so \quad (k+1)^{k+1} > (k+1)^{k+1} \times \frac{(k+1)^{k-1}}{k^k}$$

$$\therefore (k+1)^{k+1} > \frac{(k+1)^{(k+1+k-1)}}{k^k} = \frac{(k+1)^{2k}}{k^k}$$

6cii) Let $P(n)$ be the proposition that

$$n^n > (n+1)^{n-1} \text{ for } n \geq 2$$

$$P(2): 2^2 > (2+1)^{2-1},$$

$$LHS = 4, \quad RHS = 3$$

$\therefore$

$$LHS > RHS, \text{ so true for } P(2)$$

To Prove : If $P(k)$ is true then $P(k+1)$ is true for $k \geq 2$.

$$P(k): k^k > (k+1)^{k-1}$$

$$P(k+1): (k+1)^{k+1} > (k+2)^k.$$

Consider LHS of $P(k+1)$

$$\text{Assuming } P(k) \text{ true then } k^k > (k+1)^{k-1},$$

$$k > 0, \text{ then } k^k > (k+1)^{k-1}$$

$$\frac{(k+1)^{k-1}}{k^k} < 1$$

$$\text{So by i)} \quad (k+1)^{k+1} > \frac{(k+1)^{2k}}{k^k}$$

$$= \left[\frac{(k+1)^2}{k} \right]^k$$

$$= \left(\frac{k^2 + 2k + 1}{k} \right)^k$$

$$= \left(k + 2 + \frac{1}{k} \right)^k$$

$$> (k+2)^k \text{ as } k > 0$$

$\therefore P(k+1)$ true provided $P(k)$ true. Hence $P(n)$ true for integers $n \geq 2$, by Mathematical Induction.

$$7ai) I_n = \int \frac{dx}{(x^2 + 1)^n}, \quad n \geq 1$$

$$= \int \frac{1}{(x^2 + 1)^n} \frac{d}{dx}(x) dx$$

$$= \frac{x}{(x^2 + 1)^n} - (-n) \int x \cdot \frac{2x}{(x^2 + 1)^{n+1}} dx, \quad n \geq 1$$

$$= \frac{x}{(x^2 + 1)^n} + 2n \int \frac{x^2}{(x^2 + 1)^{n+1}}$$

$$= \frac{x}{(x^2 + 1)^n} + 2n \int \left[\frac{x^2 + 1}{(x^2 + 1)^{n+1}} - \frac{1}{(x^2 + 1)^{n+1}} \right] dx$$

$$= \frac{x}{(x^2 + 1)^n} + 2n \int \frac{dx}{(x^2 + 1)^n} - 2n \int \frac{dx}{(x^2 + 1)^{n+1}}$$

$$\therefore I_n = \frac{x}{(x^2 + 1)^n} + 2n I_n - 2n I_{n+1}$$

$$\therefore 2n I_{n+1} = \frac{x}{(x^2 + 1)^n} + (2n - 1) I_n$$

$$\therefore I_{n+1} = \frac{1}{2n} \left[\frac{x}{(x^2 + 1)^n} + (2n - 1) I_n \right]$$

replace n by $n-1$, for $n \geq 2$

$$I_n = \frac{1}{2(n-1)} \left[\frac{x}{(x^2 + 1)^{n-1}} + (2n-3) I_{n-1} \right]$$

Q7

$$7a) \quad I_n = \frac{1}{2(n-1)} \left\{ \left[\frac{x}{(x^2+1)^{n-1}} \right]_0^1 + (2n-3)I_{n-1} \right\}$$

$$= \frac{1}{2(n-1)} \left[\frac{1}{2} + (2n-3)I_{n-1} \right], n \geq 2$$

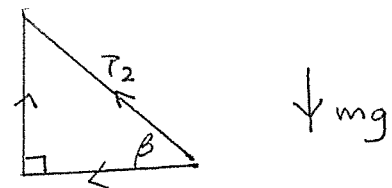
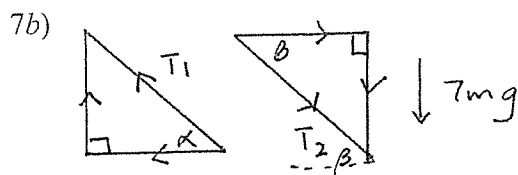
$$I_2 = \int_0^1 \frac{dx}{(x^2+1)^2}$$

$$= \frac{1}{2} \left[\frac{1}{2} + I_1 \right]$$

$$I_1 = \int_0^1 \frac{dx}{(x^2+1)} = \left[\tan^{-1} x \right]_0^1$$

$$= \tan^{-1} 1 - 0 = \frac{\pi}{4}$$

$$\therefore I_2 = \frac{1}{2} \left[\frac{1}{2} + \frac{\pi}{4} \right] = \frac{1}{4} + \frac{\pi}{8}$$



$$T_1 \cos \alpha - T_2 \cos \beta = 7maw^2 \quad (1)$$

$$T_1 \sin \alpha = T_2 \sin \beta + 7mg \quad (2)$$

$$\cos \alpha = \frac{a}{5a/3} = \frac{3}{5}$$

$$\sin \alpha = \frac{4}{5}$$

$$T_2 \cos \beta = m \cdot 2a \cdot w^2 \quad (3)$$

$$T_2 \sin \beta = mg \quad (4)$$

$$\cos \beta = \frac{a}{5a/4} = \frac{4}{5}$$

$$\sin \beta = \frac{3}{5}$$

$$7bi) \quad T_2 = \frac{mg}{\sin \beta} = mg \times \frac{5}{3} = \frac{5}{3} mg$$

bii) by (2)

$$T_1 \times \frac{4}{5} = \frac{5}{3} mg \times \frac{3}{5} + 7mg$$

$$\frac{4}{5} T_1 = 8mg$$

$$\therefore T_1 = \frac{5}{4} 8mg = 10mg$$

iii) by (1)

$$T_1 \cos \alpha - T_2 \cos \beta = \frac{7mv_1^2}{a}$$

$$10mg \times \frac{3}{5} - \frac{5}{3} mg \times \frac{4}{5} = \frac{7mv_1^2}{a}$$

$$\therefore v_1^2 = \frac{a}{7m} \left[10mg \times \frac{3}{5} - \frac{5}{3} mg \times \frac{4}{5} \right]$$

$$= \frac{ag}{7} \left[6 - \frac{4}{3} \right]$$

$$= \frac{ag}{7} \times \frac{14}{3} = \frac{2}{3} ag$$

also by (4)

$$T_2 \cos \beta = \frac{mv_2^2}{2a}$$

$$\frac{5}{3} mg \times \frac{4}{5} = \frac{m}{2a} v_2^2$$

$$v_2^2 = \frac{2a}{m} \times \frac{5}{3} mg \times \frac{4}{5}$$

$$= ag \left[2 \times \frac{5}{3} \times \frac{4}{5} \right]$$

$$= \frac{8}{3} ag$$

$$\therefore \left(\frac{v_1}{v_2} \right)^2 = \frac{2/3}{8/3} = \frac{1}{4}$$

$$\left(\frac{v_1}{v_2} \right) = \frac{1}{2}$$

Question 8

$$\begin{aligned} a) \text{ Area of slice} &= \pi(1-y)^2 \\ &= \pi(1-\sin x)^2 \end{aligned}$$

$$\text{Vol slice} = \pi(1-\sin x)^2 \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_0^{\frac{\pi}{2}} \pi(1-\sin x)^2 dx$$

$$= \pi \int_0^{\frac{\pi}{2}} 1 - 2\sin x + \frac{1}{2} - \frac{1}{2} \cos 2x \, dx \quad \left(\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x \right)$$

$$= \pi \int_0^{\frac{\pi}{2}} \left(\frac{3}{2} - 2\sin x - \frac{1}{2} \cos 2x \right) dx$$

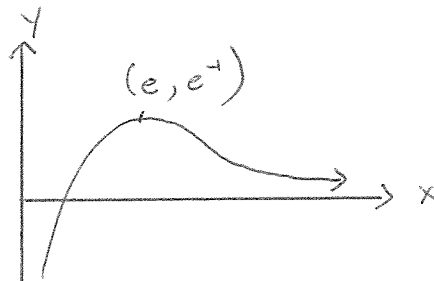
$$= \pi \left[\frac{3}{2}x + 2\cos x - \frac{1}{4}\sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left(\frac{3\pi}{4} - 2 \right)$$

$$bi) \quad h(x) = \frac{\log_e x}{x}$$

$$h'(x) = \frac{\frac{1}{x} \times x - \ln x}{x^2}$$

$$= \frac{1 - \ln x}{x^2}$$



$$bii) \text{ St.Pt. is at } x=e, y=e^{-1}$$

assuming $\pi > e$, then from the graph

$$\frac{\ln \pi}{\pi} < e^{-1} = \frac{1}{e}$$

$$\therefore \ln \pi < \frac{\pi}{e}$$

$$\therefore e^{\ln \pi} < e^{\frac{\pi}{e}}$$

$$\left(e^{\ln \pi} \right)^e < \left(e^{\frac{\pi}{e}} \right)^e$$

$$\pi^e < e^\pi$$

$$\text{ie } e^\pi > \pi^e$$

$$8ci) \frac{FD}{200-x} = \frac{10}{200}$$

$$FD = \frac{200-x}{20} = 10 - \frac{x}{20}$$

$$\therefore DC = \left(20 - \frac{x}{10}\right) \text{ as reqd.}$$

$$cii) \frac{a}{100-x} = \frac{10}{100}$$

$$a = \frac{100-x}{10}$$

$$a = 10 - \frac{x}{10}$$

$$\therefore AB = 2a = \left(20 - \frac{x}{5}\right)$$

each trap slice looks like:

Area of face is:

$$A_1 = 10 \left(20 - \frac{x}{5} + 20 - \frac{x}{20}\right)$$

$$A_1 = 400 - 3x$$

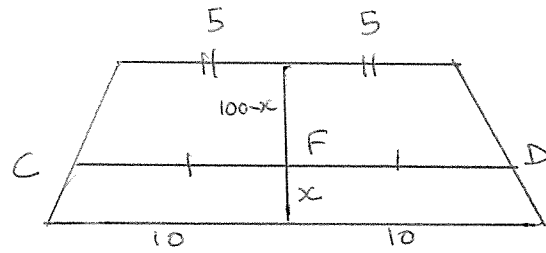
Vol of each slice is ΔV where $\Delta V = (400 - 3x)\Delta x$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \sum (400 - 3x)\Delta x$$

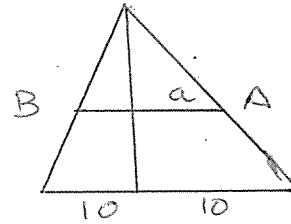
$$= \int_0^{100} (400 - 3x) dx$$

$$= \left[400x - \frac{3x^2}{2}\right]_0^{100}$$

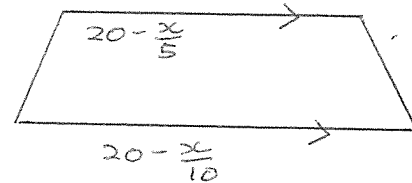
$$= 25000 \text{ m}^3$$



ground plan



roof plan



End of solutions

Marking Guidelines for THGS

Mathematics Extension 2 Trial HSC 2011

Preamble

- These guidelines were strictly implemented. Very occasionally a 'Judgment Call' may have been made. Marks were awarded, not deducted, for student's working as per guidelines.
- The 'carry through error policy' was used unless the solution was made easier by the student's error. In these cases, a 'Judgment Call' was made.
- Multiple Solutions policy. The last attempt (not crossed out) is assumed to be the actual attempt, whether or not other attempts have been crossed out. If a crossed out attempt would have gained more marks, one mark was deducted, and if higher then this became the mark.
- A red line down the left or right hand side of a script page means the content has been read by the marker.
- For each question part, only the total mark is given on the right hand side of the page, usually without comment. Sometimes working is underlined in red usually indicating that a problem with the student's working is noted.

Question 1

1(a) 2 marks

- Correct bald answer-ignore '+C'.
- Correct solution-ignore '+C'.

1 mark

- Correct integral from a correct substitution.
- Answer of the form $k\sqrt{16-x^2}$ obtained from a correct integral.

1(b) 2 marks

- Correct bald answer-ignore '+C'.
- Correct solution-ignore '+C'.

1 mark

- Correct integral with the square completed.
- Obtains answer of the form $k \tan^{-1}\left(\frac{x+2}{a}\right)$ obtained from their integral.

1(c)(i) 3 marks

- Correct answer or solution.

2 marks

- Obtains a set of answers by using a correct method.

1 mark

- Obtains a correct equation in a, b, c .

1(c)(ii) 2 marks

- Correct answer or solution.
- Obtains a correct answer from their set of values of a, b, c . (Ignore '+C'.)

1 mark

- Obtains a set of three integrals using their set of values of a, b, c .
- Obtains an answer of the form $k \ln(x^2 + 1) + l \tan^{-1} x + m \ln(x + 2)$.

1(d) 3 marks

- Correct solution-ignore '+C'.

2 marks

- Obtains an answer of the form $kx^3 \ln(x) - mx^3$, with $m > 0$, using a correct method.

1 mark

- Attempts to use correctly the method of 'Integration by Parts'.

1(e) 3 marks

- Correct solution.

2 marks

- Obtains a correct integral in terms of a polynomial in u .
- Obtains the correct answer, in terms of $\cos(x)$, from their integrand expressed as a polynomial in u .

1 mark

- Correctly substitutes $u = \cos(x)$.
- Obtains a correct answer, in terms of u , from their integrand.

Question 2**2(a)(i), (ii), (ii)****1 mark each**-correct answer.**2(b)(i)****2 marks**

- Correct answer.

1 mark

- Correct modulus.
- A correct argument.

2(b)(ii)**1 mark**

- A correct answer.
- A correct answer from their answer to 2(b)(i).

2(b)(iii)**1 mark**

- A correct answer.
- A correct answer from their answer to 2(b)(ii).

2(c)**3 marks**

- Correct diagram. (Ignore no open circle on real axis at 1, and, heavy line on all of the ray $\arg(z - 1) = \frac{3\pi}{4}$).

2 marks

- Diagram showing loci $|z - i| = 2$ and $\arg(z - 1) = \frac{3\pi}{4}$ correctly.
- Diagram with one of the two loci shown correctly with correct shading for their diagram.

1 mark

- Shading correct for their diagram with their two loci.
- One of the two loci drawn correctly.

2(d)(i)**1 mark**-correct diagram with AB parallel to OC and AB=OC marked or indicated in some way.**2(d)(ii)****1 mark**-correct answer.

2(e)

3 marks

- Correct sketch and geometrical description of locus.

2 marks

- Correct equation for locus.
- Correct sketch of locus.
- Correct geometrical description of locus.

1 mark

- Attempts to find equation of locus by a correct method.

Question 33(a)(i) 1 mark-Correct graph shape.3(a)(ii) 1 mark-Correct graph shape including cusp on x-axis.

3(a)(iii)

2marks

- Correct graph shape including the two vertical and the horizontal asymptotes.

1 mark

- Correct graph shape.
- The asymptotes $x = 1$ and $y = 0$ shown correctly on diagram.

3(a)(iv)

2 marks

- Correct graph shape including the two turning points (NOTE: gradient is indeterminate as $x \rightarrow 0^+$ from the given information, so shape ignored at $x = 0$).

1 mark

- Graph is shown to be non-negative.

3(b)(i) 1 mark – Correct graph shapes of $y = \frac{2}{x}$ and $y = \ln(x)$.

3(b)(ii)**4 marks**

- Correct graph showing correct co-ordinates of turning point, x-intercept and indication of the asymptotes.

3 marks

- Correct graph shape shown with correct co-ordinates of turning point determined.
- Correct graph shape shown with the correct two asymptotes indicated and the x-intercept given.

2 marks

- Co-ordinate of the turning point correctly determined.
- Correct two asymptotes and x-intercept indicated.

1 mark

- The horizontal asymptote is correctly determined.
- Correct graph shape is shown.

3(c)**4 marks**

- Correct graph shape shown with the following correctly indicated: the two x-intercepts, the two y-intercepts, the two asymptotes, with respective equations and common x-intercept.

3 marks

- Correct graph shape shown with the two x-intercepts and y-intercepts correctly shown.
- Correct graph shape shown with the two x-intercepts and the two asymptotes (no equations given, but common x-intercept given).

2 marks

- Correct graph shape shown with correct x-intercepts.

1 mark

- Correct graph shape shown.
- Asymptotes correctly shown (no equation or x-intercept given)
- Correct two x-intercepts and y-intercepts shown.
- Graph displays x-axis as an axis of symmetry.

Question 4**4(a)(i) 1 mark**- Correct answer.**4(a)(ii)****2 marks**

- Correct answers, or, correct answers using their value of e from 4(a)(i).

1 marks

- Obtains correct co-ordinates of foci, or, from using their value of e in 4(a)(i).
- Obtains correct x-intercepts.

4(a)(iii)**2 marks**

- Correct answers, or, correct answers using their value of e in 4(a)(i).

1mark

- Obtains correct directrices, or, correct from using their value of e in 4(a)(i).
- Obtains correct asymptotes, or, correct from using their value of e in 4(a)(i).

4(b)(i)**2 marks**

- Correct solution-must show the use of $\sin^2 \theta + \cos^2 \theta = 1$.

1 mark

- Obtains correct gradient formula in terms of θ .
- Uses $\sin^2 \theta + \cos^2 \theta = 1$ to achieve their answer.

4(b)(ii)**3 marks**

- A correct solution.

2 marks

- Obtains correct expressions for the gradients of RS and TS .

1 mark

- Attempts to use $m_1 m_2 = -1$ in their solution.

4(b)(iii) 1 mark-uses a correct method of proof.

4(c)

4 marks

- A correct solution.

3 marks

- Obtains the Cartesian equation of tangent to the ellipse and attempts to use it to find the chord of contact by a correct method.

2 marks

- Obtains Cartesian equation of tangent to the ellipse by a correct method.
- Uses a correct reasoning to find the chord of contact.

1 mark

- Obtains gradient formula for tangent at $P(x_1, y_1)$ or similar point on ellipse.
- Uses $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$ or similar result in finding equation of tangent to ellipse.
- Attempts to use correct reasoning in finding the equation of the chord of contact.

Question 5

5(a)

3 marks

- Correct solution.

2marks

- Obtains a correct equation in a, b .
- Obtains correct values of a, b from their equation(s).

1 mark

- Obtains $x^2 + 1 = (x + i)(x - i)$.
- Obtains $P(x) = (x^2 + 1)Q(x) + (x + 2)$ or similar.
- Obtains one of $P(\pm i) = \pm i + 2$ or $1 - a \pm ib$.

5(b)

2 marks

- Correct solution.

1 mark

- Obtains a correct possible graph of $y = y(x)$.
- Obtains a correct condition(s) for three real and distinct roots.

5(c)(i)

2 marks

- Correct answer in any form.

1 mark

- Obtains answer of $3! \times 4!$ or equivalent merit.
- Shows correct reasoning for permutations in a circle.

5(c)(ii)

2 marks

- Correct answer in any form.
- Correct solution using their reasoning and answer from 5(c)(i).

1 mark

- Obtains $2 \times 3! \times 3!$ as the number of arrangements.
- Shows correct reasoning for their circle permutation formula used in 5(c)(i).

5(d)(i) 1 mark – Correct answer.

5(d)(ii)

2 marks

- Correct answer.

1 mark

- Obtains $\alpha + \beta + \gamma = 2\alpha$.
- Obtains $\alpha + \beta + \gamma = \frac{1}{3}$.

5(d)(iii)

3 marks

- Correct solution.

2 marks

- Obtains p, q, r in terms of β, γ .
- Obtains $\alpha = -\frac{p}{2}$ as a root and attempts to substitute into the equation.

1 mark

- Correctly finds one of the sum of roots, sum of roots taken two at a time, product of roots.
- Correctly finds $-\frac{p}{2}$ as a root of the equation.

Question 6**6(a)(i)****2 marks**

- A correct solution.

1 mark

- Showing EC is perpendicular to AB , or equivalent merit.
- Showing $EB = AE = \frac{x}{2}$.
- Attempts to use the cosine rule.

6(a)(ii)**4 marks**

- A correct solution.

3 marks

- Obtains correctly BD in terms of x, y .
- Obtains correctly $\cos(2\alpha)$ in terms of x, y .

2 marks

- Attempts to find BD by a correct method.
- Attempts to find DC by a correct method.

1 mark

- Shows AD is perpendicular to BC .
- Shows $\angle ACB = \pi - 2\alpha$.

6(b)(i)**2 marks**

- A correct solution.

1 mark

- Shows $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^2(t) \cos^{n-2}(t) dt$.
- Shows $I_n = I_{n-2} - \frac{1}{n-1} I_n$ or equivalent merit.

6(b)(ii)

2 marks

- A correct solution.

1 mark

- Obtains $I_0 = \frac{\pi}{2}$.
- Uses the reduction formula correctly.

6(c)(i)

2 marks

- A correct solution.

1 mark

- Obtains that $(k+1)^{k+1} > (k+1)^{k+1} \times \frac{(k+1)^{k-1}}{k^k}$.
- Obtains $\frac{(k+1)^{2k}}{k^k} = (k+1)^{k+1} \times \frac{(k+1)^{k-1}}{k^k}$.

6(c)(ii)

3 marks

- A correct proof.

2 marks

- Shows statement true for $n = 2$ and uses 6(c)(i) in the inductive step or equivalent merit.

1 mark

- Shows statement true for $n = 2$.

Question 7**7(a)(i)****4 marks**

- A correct solution.

3 marks

- Obtains I_n in terms of I_{n-1} or I_{n+1} .

2 marks

- Correctly expresses $\frac{x^2}{(x^2+1)^{n+1}}$ or $\frac{1}{(x^2+1)^n}$ as two partial fractions.

1 mark

- Attempts to use integration by parts.

7(a)(ii)**2 marks**

- A correct solution.

1 mark

- Correctly uses the reduction formula for the given integral.
- Correctly obtains $\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$.

7(b) *There was a typographical error in this question. The stem was correctly worded but the diagram had mass B marked incorrectly as **m** instead of correctly as **7m**. The marking took this into account so that no student was disadvantaged. This would not have affected the outcome to 7(b)(i).*

7(b)(i)**3 marks**

- A correct solution.

2 marks

- Obtains $T_2 \sin(\beta) = mg$.

1 mark

- Obtains $\sin(\beta) = \frac{3}{5}$.
- Obtains a correct vector diagram for T_2 .

7(b)(ii)

2 marks

- A correct solution using mass at B either $7m$ or m .

1 mark

- Correctly obtains $T_1 \sin(\alpha) = T_2 \sin(\beta) + \begin{cases} 7mg \\ mg \end{cases}$ or equivalent.
- Correctly obtains T_1 from their equations.

7(b)(iii)

4 marks

- A correct solution.
- A correct solution using their results from 7(b)(i) and (ii).

3 marks

- Correctly finds v_1^2 or v_2^2 by using a correct method.

2 marks

- Correctly obtains the equations $T_1 \cos(\alpha) - T_2 \cos(\beta) = \begin{cases} \frac{7mv_1^2}{a} \\ \frac{mv_1^2}{a} \end{cases}$ and $T_2 \cos(\beta) = \frac{mv_2^2}{2a}$ or equivalent.

- Correctly obtains $\frac{v_1}{v_2}$ from their equations using their results from 7(b)(i) and (ii).

1 mark

- Shows vector force diagrams for particle B .

- Correctly obtains $T_1 \cos(\alpha) - T_2 \cos(\beta) = \begin{cases} \frac{7mv_1^2}{a} \\ \frac{mv_1^2}{a} \end{cases}$ or equivalent.

- Obtains $\cos(\beta) = \frac{4}{5}$.

Question 8**8(a)****4 marks**

- A correct solution.

3 marks

- Obtains a correct primitive from a correct integral for the volume.
- Substitutes correctly into their primitive from a correct integral for the volume.

2 marks

- Obtains a correct integral for the volume.
- Obtains correctly a volume from their integral whose integrand involves a sine function.

1 mark

- Obtains a correct integrand in their integral for the volume.
- Obtains a correct expression for their 'slice' volume.
- Obtains a correct integral from their 'slice' volume.

8(b)(i) 1 mark – A correct answer.**8(b)(ii)****2 marks**

- A correct proof.

1 mark

- Obtains correctly $\frac{\ln(\pi)}{\pi} < e^{-1}$ or equivalent merit.
- Obtains, for $x > 0$, that e^{-1} is the global maximum of $h(x)$.

8(c)(i)**2 marks**

- A correct proof.

1 mark

- Obtains a correct statement using similar triangles related to the floor of the showroom.

8(c)(ii)

6 marks

- A correct solution.

5 marks

- Obtains a correct primitive from a correct integral for the volume.

4 marks

- Obtains a correct integral for the volume.
- Obtains a correct volume from their integral.

3 marks

- Obtains a correct integrand.
- Obtains a correct primitive from their integral for the volume.

2 marks

- Obtains a correct expression for the area of the trapezium $ABCD$.

1 mark

- Obtains a correct statement using similar triangles related to the ceiling of the showroom.

END OF MARKING GUIDELINES